

28-06-2026

Agenda:

→ ML-12

→ Support vector Machines (SVM)

Support Vector Machines (SVM)

- [illegible]

Classification :-

binary or multiclass classification:

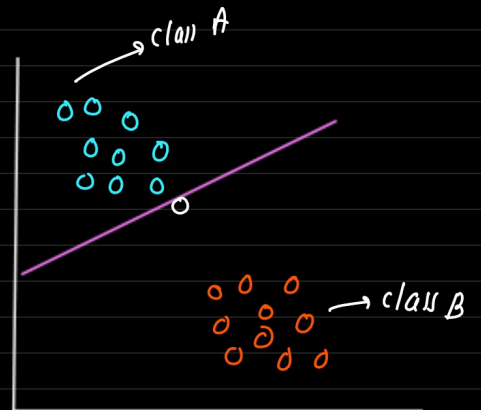
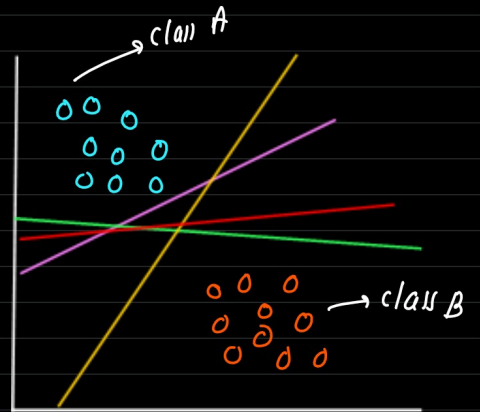
logistic regression: $\text{Blue} \rightarrow 1$ $\text{Blue} \rightarrow 1$
 $\text{green} \rightarrow 0$ $\text{green} \rightarrow 0$
svm $\text{yellow} \rightarrow 2$

data
yellow

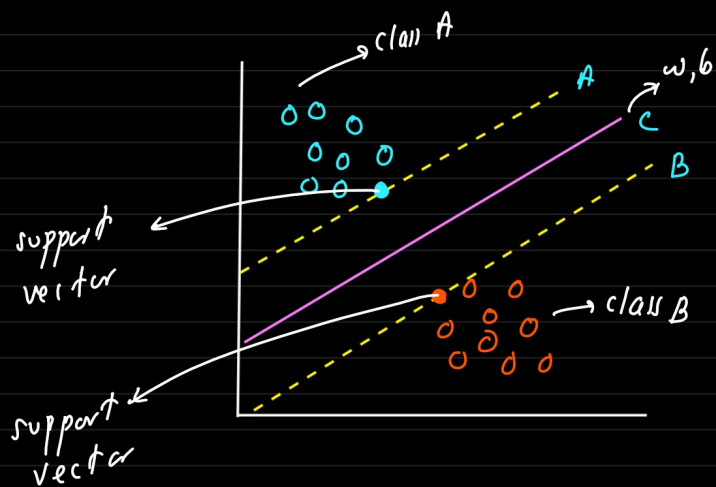
model-1 $\rightarrow$ Blue $\rightarrow$ 1 $\rightarrow$ 0.1
 $\swarrow$ other (green - yellow) $\rightarrow$ 0
 $\rightarrow$ 0.9

model-2 $\rightarrow$ green $\rightarrow 1 \rightarrow 0.2$
 $\swarrow$ other (blue, yellow) $\rightarrow 0$
 $\rightarrow 0.8$

model $\rightarrow$ yellow $\rightarrow 1 \rightarrow 0.8$
other (Blue, green) $\rightarrow 0$
 $\rightarrow 0.2$



point	x_1	x_2	class - y
A	2	7	+1 •
B	3	8	+1 •
C	4	7	+1 •
D	6	2	-1 •
E	7	3	-1 •
F	8	2	-1 •



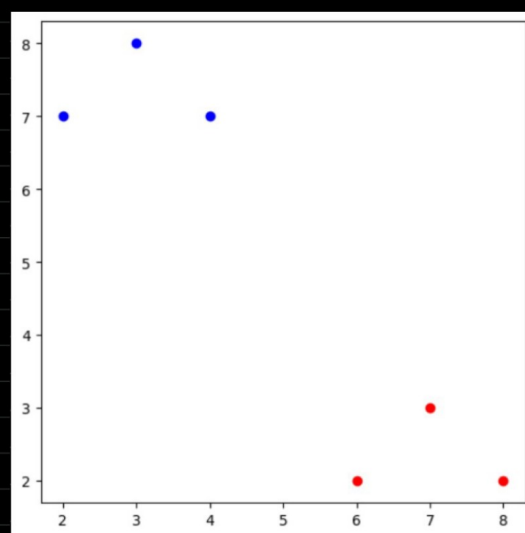
The core geometry & Math:

$$w \cdot x + b = +1 \leftarrow \text{positive class} \rightarrow A$$

$$w \cdot x + b = -1 \leftarrow \text{negative class} \rightarrow B$$

$$w \cdot x + b = 0 \leftarrow \text{center line} \rightarrow C$$

$$w_1 = 1, w_2 = 1, b = -10$$



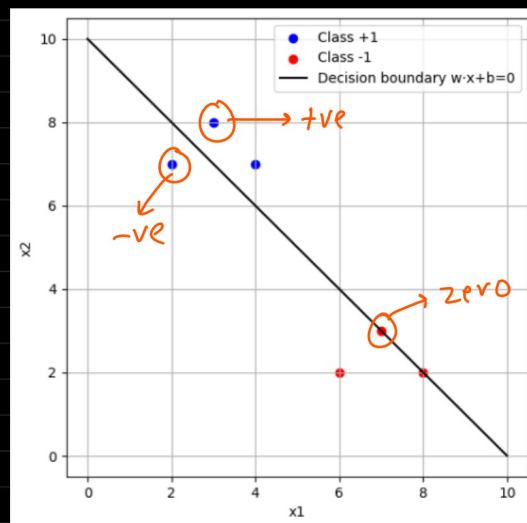
point	x_1	x_2	class - y
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$$w \cdot x + b$$

$$(1 \times 2) + (7 \times 1) - 10 = (9 - 10) = -1$$

$$(1 \times 3) + (8 \times 1) - 10 = +1$$

$$(1 \times 7) + (3 \times 1) - 10 = 0$$



Interpret:

$w \cdot x + b > 0$: point lies above the line ↗ class +1
 $w \cdot x + b < 0$: point lies below the line
 $w \cdot x + b = 0$: point is exactly on the line ↗ class -1

on the boundary

score is directly proportional to the distance of a point from the decision boundary.

$$\begin{array}{ll} A \rightarrow x_1 \rightarrow \text{score} \rightarrow +0.1 & \text{score} > 0 \\ & x_2 \rightarrow \text{score} \rightarrow +100 & \text{score} > 0 \end{array}$$

SVM doesn't just want a line where $\text{score} > 0$, it wants to fit the line to be good such that every point is at least some minimum distance away from the boundary.

positive points ($y_i = +1$) must have a score $\geq +1$

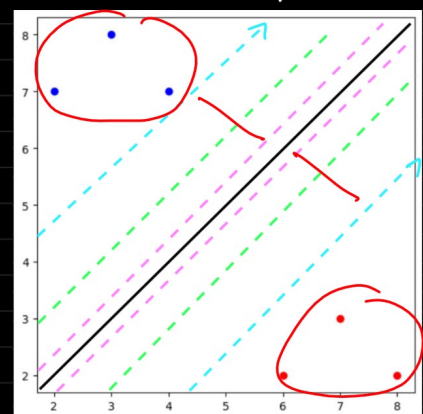
negative points ($y_i = -1$) must have a score ≤ -1

The points that are exactly at $+1, -1$ are the points that have lowest acceptable confidence. These are support vectors.

known:

require: upper-boundary $\xleftarrow{+1}$
lower-boundary $\xleftarrow{-1}$ $\longleftarrow wx + b = 0$

goal: maximum distance between two lines
margin



upper margin : $\begin{aligned} \omega \cdot x_1 + b &= 1 \\ \omega \cdot x_1 &= (1-b) \\ \omega \cdot x_1 &= c_1 \end{aligned} \quad , \quad c_1 = (1-b)$

lower margin : $\begin{aligned} \omega \cdot x_2 + b &= -1 \\ \omega \cdot x_2 &= (-1-b) \\ \omega \cdot x_2 &= c_2 \end{aligned} \quad c_2 = (-1-b)$

distance between two parallel line:

$$\text{Distance} = \frac{c_1 - c_2}{\|w\|} = \frac{(1-b) - (-1-b)}{\|w\|}$$

$$\text{margin width} = \frac{|1-b+1+b|}{\|w\|} = \frac{2}{\|w\|}$$

optimization goal: $\frac{2}{\|w\|}$ as large as possible

derivative $\rightarrow$ minimize $\frac{1}{2} \|w\|^2$

small weight equals to large margin width

The loss function:

$$\max(0, 1-z)$$

Hinge loss

$$z = y_i (\omega \cdot x_i + b)$$

the true label

the model prediction

$$y_i = +1$$

$$w \cdot x + b \rightarrow +2$$

$$\max(0, 1 - z)$$

$z \geq 1 \rightarrow$ the point is across the boundary

$$\max(0, 1 - 2)$$

$$\max(0, -1) \rightarrow 0 \text{ (No penalty)}$$

$$y_i = +1$$

$$w \cdot x + b = -2$$

$$\max(0, 1 - 2)$$

$$\max(0, 1 - +1(-2))$$

$$(0, 1 + 2) \rightarrow \max(0, 3) \rightarrow 3$$

Interpretation:

if $z \geq 1$: The point is safely across the boundary

if $z < 1$: The point is inside the margin or completely on the other side

Total SVM cost function:

$$J(w, b) = \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \max(0, 1 - y_i(w \cdot x_i + b))$$

hyperparameter $\rightarrow$ high $C \rightarrow$ skinny margin
tight / compact margin

low $C \rightarrow$ wide margin

Sub-Gradient Descent (If-Else)

$z \geq 1 \rightarrow \text{loss} \rightarrow 0$, Path A

$$\frac{\partial J}{\partial w} = w \quad (\text{only margin widening part is active})$$

$$\frac{\partial J}{\partial b} = 0$$

$(z < 1)$, loss is active/prevent, Path B

$$\frac{\partial J}{\partial w} = w - (c \cdot y_i \cdot x_i)$$

$$\frac{\partial J}{\partial b} = - (c \cdot y_i)$$

one iteration (step by step)

x	y	c	$\alpha(r)$	w	b
2.0	+1	1.0	0.1	0.4	0.0

step 1: forward pass

$$z = y \cdot (w \cdot x + b) \Rightarrow 1 (0.4 \times 2.0 + 0.0) = 0.8$$

step 2:

if $z \geq 1$

path A: gradient $\downarrow$

else

path B: gradient

$0.8 \geq 1.0$: NO $\rightarrow$ else

$$\begin{aligned} \text{gradient}_w &= w - (c \cdot y \cdot x) \\ &= 0.4 - (1.0 \cdot 1 \cdot 2.0) \\ &= -1.6 \end{aligned}$$

$$\begin{aligned} \text{gradient}_b &= -(c \cdot y) \\ &= -(1.0 \cdot 1) \\ &= -1.0 \end{aligned}$$

step 3: update weights

$$w_{\text{new}} = w_{\text{old}} - \alpha \cdot \text{gradient} \quad = 0.4 - (0.1 \cdot (-1.6)) = 0.56$$

$$b_{\text{new}} = b_{\text{old}} - \alpha \cdot \text{gradient} \quad = 0.0 - (0.1 \cdot (-1.0)) = 0.1$$

w_{old}	w_{new}	b_{old}	b_{new}
0.4	0.56	0.0	0.1

$$y(w \cdot x + b) \rightarrow 0.8$$

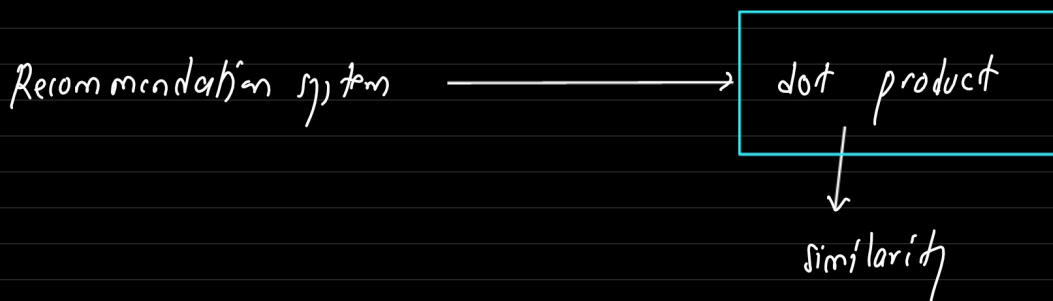
$$\rightarrow +1 ((0.56 \cdot 2.0) + 0.1) = 1.22$$

$$y_{\text{actual}} \rightarrow +1$$

$$y_{\text{actual}} \rightarrow +1$$

Algo of learning

- (1) forward pass
 - (2) gradient
 - (3) update weight
 - (4) pred + error
 - (5) repeat until converged ($1 - \epsilon$)
-



Movie recommendation system

	Action	Comedy	
User A	$[2]$	$[0]$	$\rightarrow$ Fast & Furious
User B	10	0	$\rightarrow$ John Wick, Pushpa, Expando, MI
User C	$[0]$	$[10]$	

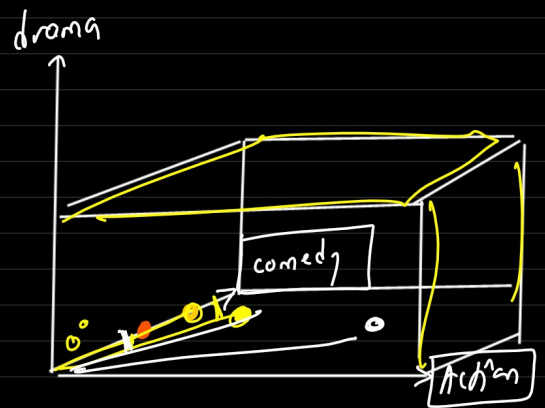
$$A \rightarrow B \rightarrow \text{dot product} \rightarrow (2 \times 10) + (0 \times 0) = 20$$
$$A \rightarrow C \rightarrow \text{dot product} \rightarrow (2 \times 0) + (0 \times 10) = 0$$

$$\cos(\text{angle}) = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \times \|\vec{b}\|}$$

$$A[2,0] = \sqrt{2^2 + 0^2} = 2$$

$$B[10,0] = \sqrt{10^2 + 0^2} = 10$$

$$= \frac{2 \cdot 10}{2 \times 10} = \frac{20}{20} = 1.0$$



Dot product $\rightarrow 20 \rightarrow B$ likes action more

Cosine similarity $\rightarrow 1.0 \rightarrow$ share some taste or not

Amazon - recm. $\rightarrow$ smartpho

A (_ _ _ _)

$\rightarrow$ sma.

$\rightarrow$ cur

$\rightarrow$ ch

B (_ _ _ _)

$\rightarrow$ smart

$\rightarrow$ _

C ()

laptop
game

